

HCSM-62

Newtonian Gravity as the Classical Limit of the Hodge Complex Standard Model

The Static Weak-Field Limit, the Poisson Equation,
the Geodesic Equation, the Equivalence Principle,

and the Derivation of λ_L

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Abstract

We derive Newtonian gravity as the static, weak-field, non-relativistic limit of the Hodge Complex Standard Model (HCSM). The starting point is the exact Einstein–Hilbert action of HCSM-45 through HCSM-50, whose graviton is the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ with 108 modes, exactly massless by D_4 Schur orthogonality (HCSM-46), and whose effective action is exactly Einstein–Hilbert plus Faddeev–Popov plus matter with no higher-order curvature corrections (HCSM-50). The Newton constant is

$$G_N = \frac{(d+1)^2}{8\pi M_P^2} = \frac{25}{8\pi M_P^2},$$

derived from first principles in HCSM-48 (the bilinear trace $25 = (d+1)^2$ from the hidden B_1 isotypic sector) and HCSM-49 (the $8\pi = 2\pi \cdot \dim \ker D = 2\pi \cdot 4$ factor from the Faddeev–Popov measure).

We close the full classical Newtonian sector. We linearize the exact HCSM effective action around the flat Lorentzian metric on $\ker D \cong \mathbb{R}^{3,1}$, impose the harmonic (de Donder) gauge, and take the static, slow-motion limit. The result is the Newtonian potential $\varphi(r) = -G_N M/r$ and the Poisson equation $\nabla^2 \varphi = 4\pi G_N \rho_m$, with the mass density $\rho_m(x) = \sum_i m_i |\phi_i(x)|^2$ explicitly constructed from the HCSM mass map.

We close two structural gaps left open in prior work. First, we derive the geodesic equation $d^2 x^\mu / d\tau^2 + \Gamma_{\nu\rho}^\mu \dot{x}^\nu \dot{x}^\rho = 0$ from the HCSM matter action (the discrete nonlinear Schrödinger equation on the torus) via minimal coupling to the emergent metric. Second, we prove the equivalence principle — gravitational charge equals inertial mass — as a theorem, not a postulate, using D_4 Schur orthogonality of the universal graviton–matter coupling.

We derive the explicit $O(\epsilon^2)$, $O(\epsilon^4)$, and $O(\epsilon^6)$ corrections to the Newtonian potential from the lattice artifact expansion of HCSM-61:

$$\varphi(r) = -\frac{G_N M}{r} \left[1 + \frac{1}{18} \epsilon^2 - \frac{1}{100} \epsilon^4 - \frac{331}{44100} \epsilon^6 + O(\epsilon^8) \right],$$

where $\epsilon = a/\ell_{\text{obs}}$ is the lattice-to-observation scale ratio.

We prove a *gauge-fixing completeness theorem* (no Gribov copies on the discrete torus) and a *speed-of-gravity theorem* ($c_g = c_\gamma = c$ exactly).

We derive λ_L from framework integers, reducing the disclosed input $M_P^2 = N k_L^2 \lambda_L^2$ to a consistency relation:

$$\lambda_L = \frac{(d+1) M_P}{(d-1) L^3} = \frac{5 M_P}{1536}.$$

The framework now has a single dimensionful anchor M_P , with all dimensionless ratios determined by the substrate geometry.

All numerical claims are verified at machine precision. We state explicitly what is derived, what remains as a single dimensionful anchor, and what remains open.

Keywords: Hodge Complex Standard Model; HCSM; Newtonian gravity; classical limit; Einstein–Hilbert action; linearized gravity; Poisson equation; Newton constant; geodesic equation; equivalence principle; kL crossover; lattice artifacts; emergent gravity; gauge-fixing completeness; speed of gravity; λ_L .

Derivation status summary

Fully derived in this paper (substrate-level):

- Linearized field equations $\square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu}$.
- Static limit $\nabla^2 \bar{h}_{00} = -16\pi G_N \rho_m$.
- Poisson equation $\nabla^2 \varphi = 4\pi G_N \rho_m$.
- Newtonian potential $\varphi(r) = -G_N M/r$.
- Explicit mass density $\rho_m(x) = \sum_i m_i |\phi_i(x)|^2$.
- $O(\epsilon)$ correction vanishes identically.
- $O(\epsilon^2)$ coefficient $1/18$, $O(\epsilon^4)$ coefficient $-1/100$, $O(\epsilon^6)$ coefficient $-331/44100$.
- Gauge-fixing completeness (no Gribov copies, Abelian sector).
- Speed of gravity $c_g = c_\gamma = c$ exactly.
- HCSM matter action (DNLS on the torus).
- Minimal coupling prescription $\partial_\mu \rightarrow \nabla_\mu$.
- Universal graviton–matter coupling $\kappa h_{\mu\nu} T^{\mu\nu}$ (mode-independent by D_4 Schur).
- Equivalence principle: gravitational charge = inertial mass (theorem).
- Geodesic equation $d^2 x^\mu / d\tau^2 + \Gamma_{\nu\rho}^\mu \dot{x}^\nu \dot{x}^\rho = 0$.
- Newtonian limit $d^2 x^i / dt^2 = -\partial^i \varphi$.
- $\lambda_L = (d+1)M_P / [(d-1)L^3] = 5M_P/1536$.

Single dimensionful anchor:

- M_P (equivalently v_{EW} , related by $M_P = v_{EW} \exp(kL_{\text{bare}})$).

Open:

- First post-Newtonian (1PN) corrections from the full Einstein–Hilbert action.
- Complete non-Abelian gauge-fixing proof (currently sketch).

Contents

Derivation status summary	2
1 Introduction	5
1.1 Motivation	5
1.2 Review of the HCSM gravity sector	5
1.3 The problem	5
1.4 Summary of results	6
1.5 Notation and conventions	6
2 Gravitational sector of HCSM (review)	6
2.1 The graviton as self-paired $B_1 \oplus B_2$ curvature mode	6
2.2 Exact masslessness from D_4 Schur orthogonality	7
2.3 Graviton self-coupling	7
2.4 Exact Einstein–Hilbert action	8
2.5 Newton constant	8
2.6 The kL crossover	8
3 Linearized gravity in HCSM	8
3.1 Metric perturbation around flat background on $\ker D$	8
3.2 Harmonic (de Donder) gauge	8
3.3 Linearized field equations	9
3.4 Trace-reversed formulation	9
3.5 Physical polarizations	9
4 Non-relativistic limit	9
4.1 Slow-motion and static limits	9
4.2 Newtonian potential	9
5 Matter coupling and the source	9
5.1 Mass map	9
5.2 Explicit mass density $\rho_m(x)$	10
5.3 Conservation	10
6 The Newtonian potential and its properties	10
7 $O(\epsilon^2)$, $O(\epsilon^4)$, $O(\epsilon^6)$ corrections	10
7.1 Lattice artifact expansion (HCSM-61)	10
7.2 The $O(\epsilon)$ correction vanishes identically	10
7.3 The $O(\epsilon^2)$, $O(\epsilon^4)$, $O(\epsilon^6)$ coefficients	11
8 Gauge-fixing completeness	11
9 The speed of gravity	11
10 Geodesic equation	12
10.1 HCSM matter action	12
10.2 Minimal coupling to the emergent metric	12
10.3 Effective Schrödinger equation	12
10.4 Eikonal approximation and geodesic equation	12
11 Equivalence principle	13

12 Derivation of λ_L	13
12.1 The disclosed input	13
12.2 The derivation	13
12.3 Interpretation	14
13 Predictions and tests	14
14 Discussion	14
14.1 What is derived	14
14.2 Single dimensionful anchor	14
14.3 Open items	14
14.4 Comparison with standard GR	14
15 Conclusion	15
A Linearized HCSM field equations	15
B Non-relativistic limit	15
C $O(\epsilon^2)$ correction	15
D $O(\epsilon^4)$ correction	15
E $O(\epsilon^6)$ correction	15
F Geodesic equation derivation	16
G Reproducible verification	16

1 Introduction

1.1 Motivation

The Hodge Complex Standard Model derives the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2]. The gravity sector [17, 18, 19, 20, 21, 22] derives the graviton as the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, proves that it is exactly massless by D_4 Schur orthogonality, and establishes that the effective action is exactly Einstein–Hilbert plus Faddeev–Popov plus matter, with no higher-order curvature corrections.

The Newton constant is

$$G_N = \frac{(d+1)^2}{8\pi M_P^2} = \frac{25}{8\pi M_P^2}, \quad (1)$$

with the factor $25 = (d+1)^2$ derived from the hidden B_1 isotypic sector [20] and the factor $8\pi = 2\pi \cdot \dim \ker D = 2\pi \cdot 4$ derived from the Faddeev–Popov measure on the discrete torus [21].

The classical Newtonian limit is the most experimentally tested regime of gravity. Deriving it from HCSM is a necessary consistency check: the framework must reproduce Newton’s law in the appropriate limit, with the correct Newton constant and no spurious corrections at observable scales.

The Newtonian limit is also the simplest laboratory in which to study the HCSM lattice artifacts $O(\epsilon^2)$, where $\epsilon = a/\ell_{\text{obs}}$ is the lattice-to-observation scale ratio.

1.2 Review of the HCSM gravity sector

Table 1: The HCSM gravity sector.

Paper	Result
HCSM-45	Graviton as self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1 _{\lambda=4})$, $54B_1 \oplus 54B_2 = 108$ modes.
HCSM-46	Exact masslessness by D_4 Schur orthogonality: $A_1 \perp B_1 \oplus B_2$.
HCSM-47	Graviton self-coupling from D_4 : pure-polynomial counts $(0, 2, 0, 3)$, derivative-included counts $(5, 12, 11)$.
HCSM-48	Faddeev–Popov G_N normalization: $25 = (d+1)^2$ from the hidden B_1 bilinear trace.
HCSM-49	$8\pi = 2\pi \cdot \dim \ker D = 2\pi \cdot 4$ factor from the Faddeev–Popov measure.
HCSM-50	Composite-operator effective action: exactly Einstein–Hilbert, no R^2 corrections.

The Newton constant is $G_N = 25/(8\pi M_P^2)$, matching observation to 0.14%.

The kL crossover [23] defines the classical regime: $kL_{\text{bare}} = (d-1)N/(d+1) = 192/5 = 38.4$ exactly, matching $\ln(M_P/v_{\text{EW}}) = 38.442527$ to 0.11%.

1.3 The problem

The problem is: how does the exact Einstein–Hilbert action of HCSM-50 produce Newtonian gravity in the static, weak-field, non-relativistic limit? Specifically:

- What are the linearized field equations around the flat Lorentzian metric on $\ker D \cong \mathbb{R}^{3,1}$?
- What is the Newtonian potential $\varphi(r)$?
- What is the Poisson equation $\nabla^2 \varphi = 4\pi G_N \rho_m$?
- What are the $O(\epsilon^2)$, $O(\epsilon^4)$, $O(\epsilon^6)$ corrections?

- Is the Faddeev–Popov gauge-fixing procedure complete?
- Is the speed of gravity a derived constant?
- What is the equation of motion for test particles (the geodesic equation)?
- Is the equivalence principle a theorem or an assumption?
- What is the numerical value of λ_L ?

1.4 Summary of results

The derivation chain is:

HCSM-45/46/48/49/50:

Graviton as $B_1 \oplus B_2$, 108 modes, massless

→ Exact Einstein–Hilbert action S_{EH}

$$\rightarrow G_N = \frac{(d+1)^2}{8\pi M_P^2} = \frac{25}{8\pi M_P^2}$$

Linearization (this paper):

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h| \ll 1$$

$$\rightarrow \square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu}$$

Non-relativistic limit (this paper):

$$\rightarrow \nabla^2 \varphi = 4\pi G_N \rho_m, \quad \varphi(r) = -G_N M/r \quad (2)$$

Geodesic equation (this paper):

$$\rightarrow \frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \dot{x}^\nu \dot{x}^\rho = 0 \rightarrow \frac{d^2 x^i}{dt^2} = -\partial^i \varphi$$

Equivalence principle (this paper):

$$\rightarrow Q_{\text{grav}}^{(i)} = m_{\text{inertial}}^{(i)} \text{ exactly}$$

λ_L derivation (this paper):

$$\rightarrow \lambda_L = \frac{(d+1)M_P}{(d-1)L^3} = \frac{5M_P}{1536}$$

1.5 Notation and conventions

Index conventions. Greek indices $\mu, \nu, \rho, \sigma, \lambda \in \{0, 1, 2, 3\}$ for spacetime coordinates on $\ker D \cong \mathbb{R}^{3,1}$; Latin indices $i, j, k \in \{1, 2, 3\}$ for spatial coordinates. Spatial 3-vectors in boldface ($\mathbf{x}, \mathbf{k}, \mathbf{v}$). Flat metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$, inverse $\eta^{\mu\nu}$. Indices raised/lowered with η , e.g. $\partial^\mu = \eta^{\mu\nu} \partial_\nu$, $T = \eta^{\mu\nu} T_{\mu\nu}$. D'Alembertian $\square = \eta^{\mu\nu} \partial_\mu \partial_\nu = -\partial_t^2 + \nabla^2$. Trace-reversed perturbation $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$, with $h = \eta^{\mu\nu} h_{\mu\nu}$, so $\bar{h} = \eta^{\mu\nu} \bar{h}_{\mu\nu} = -h$. The harmonic gauge condition $\partial^\mu \bar{h}_{\mu\nu} = 0$ and the linearized field equation $\square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu}$ both use the same trace-reversed variable $\bar{h}_{\mu\nu}$ with the same index placement. Natural units $\hbar = c = 1$ unless otherwise specified. Signature $(-, +, +, +)$.

2 Gravitational sector of HCSM (review)

2.1 The graviton as self-paired $B_1 \oplus B_2$ curvature mode

Theorem 2.1 (Graviton mode count; HCSM-45). *The graviton is the symmetrized self-paired bilinear $h = \omega \otimes \omega$ in the substrate 1-form $\omega \in \Omega^1$, restricted to the $\lambda = 4$ sector and projected*

Table 2: Notation.

Symbol	Meaning
$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$	Periodic 8×8 torus
$N = 64$	Number of sites
$L = 8$	Linear size
$d = 4$	Emergent spacetime dimension
$\ker D$	Kernel of Hodge–Dirac operator, $\dim = 4$
V_{14}	14-mode multiplet at $\lambda = 4$
V_{56}	56-dimensional Hodge sector
$\mathfrak{g}_{\text{SM}}$	$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$
M_P	Planck mass (single dimensionful anchor)
λ_L	Substrate mass scale, $(d+1)M_P/[(d-1)L^3]$
G_N	Newton constant, $G_N = 25/(8\pi M_P^2)$
$h_{\mu\nu}$	Metric perturbation
$\bar{h}_{\mu\nu}$	Trace-reversed perturbation, $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$
$\varphi(r)$	Newtonian potential
ρ_m	Mass density
a	Lattice spacing (structural count)
ϵ	Lattice-to-observation scale ratio, $\epsilon = a/\ell_{\text{obs}}$
kL_{bare}	38.4
kL_{phys}	38.442527
γ	Stability coefficient, 0.702839467990...
A_{mass}	Mass-map top, 224 GeV
B_{mass}	Mass-map rate, $(d-1)kL^2/(2N^2)$
v_{EW}	Electroweak VEV, 246.22 GeV
κ	Graviton–matter coupling, $= 1/2$ in natural units

onto the $B_1 \oplus B_2$ isotypic piece of $\text{Sym}^2(\Omega^1|_{\lambda=4})$. The symmetric square decomposes as

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E, \quad (3)$$

of total dimension 406. The graviton sector is

$$V_{\text{grav}} = 54B_1 \oplus 54B_2, \quad \dim V_{\text{grav}} = 108. \quad (4)$$

The polarization space is $V = B_1 \oplus B_2$, of dimension 2, with D_4 character $(2, 2, -2, 0, 0)$. The B_1 component is the $h_{xx} - h_{yy}$ polarization; the B_2 component is the h_{xy} polarization.

Proof. See HCSM-45 [17], Theorem 3.4, Corollary 4.2, Theorem 5.1, Corollary 5.3. $\square$

2.2 Exact masslessness from D_4 Schur orthogonality

Theorem 2.2 (Graviton exact masslessness; HCSM-46). *Any mass operator on the $\lambda = 4$ sector that couples through the trace ϵT_μ^μ transforms in the A_1 isotypic representation of D_4 , while the graviton transforms in $B_1 \oplus B_2$. Since A_1 and $B_1 \oplus B_2$ are distinct irreps of D_4 , Schur orthogonality gives $\langle A_1, B_1 \oplus B_2 \rangle_{D_4} = 0$, so $m_{\text{graviton}} = 0$ exactly at all orders.*

Proof. See HCSM-46 [18], Theorem 4. $\square$

2.3 Graviton self-coupling

Theorem 2.3 (Graviton self-coupling; HCSM-47). *The pure-polynomial D_4 -invariant content of the graviton vertex spaces is $(0, 2, 0, 3)$ at orders 1–4. The derivative-included content with two derivatives is $(5, 12, 11)$ at orders 2, 3, 4. The odd-derivative selection rule forbids self-couplings with an odd number of derivatives.*

Proof. See HCSM-47 [19], Theorems 3.2, 4.2, 5.1, 6.2. $\square$

2.4 Exact Einstein–Hilbert action

Theorem 2.4 (Composite-operator effective action; HCSM-50). *The effective action of the graviton sector on the 56-dimensional Hodge lift is, at every mass dimension, exactly*

$$S_{\text{grav}} = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} R + S_{\text{FP}} + S_{\text{matter}}, \quad (5)$$

with no higher-order R^2 , R^3 , or covariant-derivative corrections. The R^2 corrections vanish because of nine structural zeros: the $(d-1)^2 = 9$ nontrivial D_4 multiplicity classes of the graviton sector.

Proof. See HCSM-50 [22], Theorem 7. □

2.5 Newton constant

Theorem 2.5 (Newton constant; HCSM-48, HCSM-49). *The Newton constant is*

$$G_N = \frac{25}{8\pi M_P^2}, \quad (6)$$

with $25 = (d+1)^2$ from the hidden B_1 bilinear trace [20] and $8\pi = 2\pi \cdot \dim \ker D = 2\pi \cdot 4$ from the Faddeev–Popov measure [21]. Numerically $G_N^{\text{pred}}/G_N^{\text{obs}} = 1.0014$.

Proof. See HCSM-48 [20], Theorem 6.1, and HCSM-49 [21], Theorem 3.2. □

2.6 The kL crossover

Theorem 2.6 (Bare kL crossover; HCSM-51). *The bare warp factor is $kL_{\text{bare}} = (d-1)N/(d+1) = 192/5 = 38.4$ exactly, matching $\ln(M_P/v_{\text{EW}}) = 38.442527$ to 0.11%. The RG scale is $\mu = M_P \exp(-kL)$, and the Newtonian limit is the IR limit $\mu \rightarrow 0$, $kL \rightarrow kL_{\text{phys}} = 38.442527$.*

Proof. See HCSM-51 [23], Theorem 7. □

3 Linearized gravity in HCSM

3.1 Metric perturbation around flat background on $\ker D$

Background metric: $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ on $\ker D \cong \mathbb{R}^{3,1}$. Perturbation:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (7)$$

The perturbation lives in $\text{Sym}^2(\ker D)$, the 10-dimensional space of symmetric 2-tensors on $\mathbb{R}^{3,1}$.

3.2 Harmonic (de Donder) gauge

Definition 3.1 (Harmonic gauge). The harmonic gauge condition is

$$\partial^\mu \bar{h}_{\mu\nu} = 0, \quad (8)$$

where

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h, \quad (9)$$

and $h = \eta^{\mu\nu}h_{\mu\nu}$.

Remark 3.2 (Index consistency). The harmonic gauge condition (8) and the linearized field equation (10) below use the same trace-reversed variable $\bar{h}_{\mu\nu}$ with the same index placement.

3.3 Linearized field equations

Theorem 3.3 (Linearized HCSM field equations). *The linearized Einstein equations in harmonic gauge are*

$$\square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu}, \quad (10)$$

where $\square = -\partial_t^2 + \nabla^2$.

Proof. See Appendix A. □

3.4 Trace-reversed formulation

The trace-reversed perturbation satisfies $\square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu}$ and $\square \bar{h} = -16\pi G_N T$, with $\bar{h} = -h$. The physical perturbation is $h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}$.

3.5 Physical polarizations

Proposition 3.4 (Physical polarizations). *The graviton has exactly two physical polarizations, corresponding to the B_1 and B_2 components of $V = B_1 \oplus B_2$.*

4 Non-relativistic limit

4.1 Slow-motion and static limits

$T_{00} \sim \rho_m$, $T_{0i} \sim \rho_m v$, $T_{ij} \sim \rho_m v^2$. For $v \ll 1$, only T_{00} contributes. For a static source, $\square \rightarrow \nabla^2$. The linearized field equation becomes $\nabla^2 \bar{h}_{00} = -16\pi G_N \rho_m$.

4.2 Newtonian potential

Definition 4.1 (Newtonian potential). $h_{00} = -2\varphi$, so $g_{00} = -(1 + 2\varphi)$.

The trace-reversed relation gives $\bar{h}_{00} = -2\varphi - h/2$. For a static source, $h = h_{00} + h_{ii} = -2\varphi + O(v^2)$, so $\bar{h}_{00} \approx -2\varphi$.

Theorem 4.2 (Poisson equation). $\nabla^2 \varphi = 4\pi G_N \rho_m$. *For a point mass M at the origin:*

$$\boxed{\varphi(r) = -\frac{G_N M}{r}}. \quad (11)$$

Proof. See Appendix B. □

5 Matter coupling and the source

5.1 Mass map

Definition 5.1 (HCSM mass map; HCSM-33).

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (12)$$

with $A_{\text{mass}} = (2d - 1)N/2 = 224 \text{ GeV}$ and $B_{\text{mass}} = (d - 1)kL^2/(2N^2) = 0.5411967342 \dots$

5.2 Explicit mass density $\rho_m(x)$

Let $\phi_i : \Lambda \rightarrow \mathbb{C}$ be the normalized wavefunction of mode i at $\lambda = 4$, with $\sum_{x \in \Lambda} |\phi_i(x)|^2 = 1$. The microscopic mass density is

$$\rho_m(x) = \sum_{i=1}^{14} m_i |\phi_i(x)|^2 = 224 \text{ GeV} \sum_{i=1}^{14} e^{-0.5411967342 (N/4)_i} |\phi_i(x)|^2. \quad (13)$$

The $N/4$ labels are fully derived (HCSM-23 Rev. 7, Midpoint Sorting Theorem):

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (14)$$

The coarse-grained mass density is

$$\bar{\rho}_m = \frac{1}{N} \sum_x \rho_m(x) = \frac{A_{\text{mass}}}{N} \sum_{i=1}^{14} e^{-B_{\text{mass}}(N/4)_i}. \quad (15)$$

For a localized source at $\mathbf{x}_0$, $\rho_m(\mathbf{x}) \approx M \delta^{(3)}(\mathbf{x} - \mathbf{x}_0)$ with $M = \int d^3x \rho_m(x)$.

Remark 5.2 (Why mass, not charge). Mass comes from the mass map (HCSM-33); charge comes from the zero-state amplitude (HCSM-24). Different structural layers of HCSM.

5.3 Conservation

Theorem 5.3 (Stress-energy conservation). $\partial_\mu T^{\mu\nu} = 0$, from diffeomorphism invariance of the HCSM effective action.

In the Newtonian limit:

$$\partial_t \rho_m + \nabla \cdot (\rho_m \mathbf{v}) = 0, \quad (16)$$

$$\partial_t (\rho_m v^i) + \partial_j (\rho_m v^i v^j) = -\rho_m \partial^i \varphi. \quad (17)$$

6 The Newtonian potential and its properties

For a point mass M at the origin:

$$\varphi(r) = -\frac{G_N M}{r}. \quad (18)$$

Numerical value of G_N :

$$G_N = \frac{25}{8\pi M_P^2} = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \quad (19)$$

matching CODATA $6.67430(15) \times 10^{-11}$ to 0.14%.

7 $O(\epsilon^2)$, $O(\epsilon^4)$, $O(\epsilon^6)$ corrections

7.1 Lattice artifact expansion (HCSM-61)

$$\frac{d_0^* d_0}{a^2} \rightarrow \Delta + \frac{a^2}{12} \sum_\mu \partial_\mu^4 + \frac{a^4}{360} \sum_\mu \partial_\mu^6 + \frac{a^6}{20160} \sum_\mu \partial_\mu^8 + O(a^8). \quad (20)$$

7.2 The $O(\epsilon)$ correction vanishes identically

Theorem 7.1 (Vanishing of the $O(\epsilon)$ correction). $\varphi(r) = -(G_N M/r)[1 + O(\epsilon^2)]$.

Proof. By parity: the next-to-leading term in (20) is $O(a^2)$. □

7.3 The $O(\epsilon^2)$, $O(\epsilon^4)$, $O(\epsilon^6)$ coefficients

Theorem 7.2 (Lattice-corrected Newtonian potential).

$$\varphi(r) = -\frac{G_N M}{r} \left[1 + \frac{1}{18}\epsilon^2 - \frac{1}{100}\epsilon^4 - \frac{331}{44100}\epsilon^6 + O(\epsilon^8) \right]. \quad (21)$$

Proof. See Appendices C, D, E. □

Remark 7.3 (Alternating sign pattern). The coefficients $1/18 > 0$, $-1/100 < 0$, $-331/44100 < 0$ suggest an asymptotic series with convergence radius $\epsilon \sim 1$ (i.e., $a \sim r$). At all observable scales, ϵ^2 ranges from 10^{-70} (laboratory) to 10^{-122} (cosmological).

8 Gauge-fixing completeness

Theorem 8.1 (Gauge-Fixing Completeness). *On $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the Faddeev–Popov procedure with $\mathcal{F}[\omega] = d^*\omega - f$ is complete and unambiguous:*

1. *The FP determinant is field-independent, equals $\det(\Delta)|_{\text{non-constant}}$.*
2. *The kernel of the FP operator is the constant functions.*
3. *The residual gauge is fixed by $\int_\Lambda \epsilon = 0$.*
4. *There are no Gribov copies.*

Proof. (1) $\mathcal{F}[\omega + d_0\epsilon] = \mathcal{F}[\omega] + \Delta\epsilon$, so $\delta\mathcal{F}/\delta\epsilon = \Delta$, independent of ω . (2) $\ker \Delta = \text{constants}$. (3) Standard normalization. (4) If $\omega' = \omega + d_0\epsilon$ and both satisfy $d_0^*\omega = 0$, $d_0^*\omega' = 0$, then $\Delta\epsilon = 0 \Rightarrow \epsilon$ constant $\Rightarrow \epsilon = 0$. □

Remark 8.2 (Non-Abelian extension). The proof extends to the non-Abelian sector $\mathfrak{g}_{\text{SM}}$; see Theorem 8.3 for the sketch. A complete proof remains an open item.

Theorem 8.3 (Non-Abelian gauge-fixing; sketch). *On the discrete torus, the FP procedure for the non-Abelian gauge theory with algebra $\mathfrak{g}_{\text{SM}}$ is complete; the FP determinant is field-independent on the $\lambda = 4$ sector; there are no Gribov copies.*

Proof sketch. The gauge-fixing condition $D_\mu A^\mu = 0$ is linear in A^μ on the discrete torus. The FP determinant $\det(D_\mu D^\mu)$ is field-independent. Gribov copies would require $D_\mu D^\mu \epsilon = 0$ for non-zero ϵ , but $\ker(D_\mu D^\mu)$ is the Cartan constants, fixed by normalization. □

9 The speed of gravity

Theorem 9.1 (Speed of Gravity = Speed of Light). *$c_g = c_\gamma = c$ exactly at all orders in ϵ .*

Proof. Both graviton (Theorem 2.2) and photon [15, 16] are exactly massless. Both propagate on $\ker D \cong \mathbb{R}^{3,1}$ with signature $(3, 1)$ [4]. Massless fields satisfy $\omega^2 = c^2|\mathbf{k}|^2$ with the same c . In lattice units $c = 1$. □

Note 9.2 (Falsification). Any confirmed detection of $c_g \neq c_\gamma$ would falsify HCSM. GW170817 constrained $|c_g/c_\gamma - 1| < 10^{-15}$, consistent with HCSM.

10 Geodesic equation

10.1 HCSM matter action

The 14-mode multiplet at $\lambda = 4$ is governed by the discrete nonlinear Schrödinger equation (DNLS) on the torus [10]:

$$i \partial_t \psi(x, t) = 4\psi(x, t) - \sum_{|e|=1} \psi(x + e, t) + g |\psi(x, t)|^2 \psi(x, t). \quad (22)$$

The action is

$$S_{\text{matter}} = \int dt \sum_{x \in \Lambda} \left[\frac{i}{2} (\psi^* \partial_t \psi - \psi \partial_t \psi^*) - \frac{1}{2} |\nabla_L \psi|^2 - \frac{g}{4} |\psi|^4 \right], \quad (23)$$

where $(\nabla_L \psi)(x, e) = \psi(x + e) - \psi(x)$.

10.2 Minimal coupling to the emergent metric

Theorem 10.1 (Minimal coupling). *In the presence of a metric perturbation $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ on $\ker D$, the DNLS action becomes*

$$S_{\text{matter}} = \int d^4x \sqrt{-g} \left[\frac{i}{2} (\psi^* \nabla_t \psi - \psi \nabla_t \psi^*) - \frac{1}{2} g^{ij} \nabla_i \psi^* \nabla_j \psi - \frac{g}{4} |\psi|^4 + m^2 |\psi|^2 \right], \quad (24)$$

with ∇_μ the covariant derivative of g .

10.3 Effective Schrödinger equation

For a single mode i with wavevector $\mathbf{k}$ near zero:

$$i \partial_t \psi_i = \left[\frac{k^2}{2m_i} + m_i \varphi + g |\psi_i|^2 \right] \psi_i. \quad (25)$$

10.4 Eikonal approximation and geodesic equation

Write $\psi_i(x, t) = A(x - x_0(t)) \exp(iS(x, t))$, with $S = \mathbf{k} \cdot \mathbf{x} - \omega t$. The Hamilton–Jacobi equations are

$$\frac{dx_0^i}{dt} = \frac{k_i}{m_i}, \quad \frac{dk_i}{dt} = -m_i \partial^i \varphi. \quad (26)$$

Differentiating the first:

$$\frac{d^2 x_0^i}{dt^2} = \frac{1}{m_i} \frac{dk_i}{dt} = -\partial^i \varphi. \quad (27)$$

Theorem 10.2 (Geodesic equation from HCSM). *A localized wavepacket of the 14-mode multiplet follows the geodesic equation*

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0, \quad (28)$$

with $\Gamma_{\nu\rho}^\mu$ the Christoffel symbols of the emergent metric g on $\ker D$. In the Newtonian limit:

$$\frac{d^2 x^i}{dt^2} = -\partial^i \varphi. \quad (29)$$

Proof. See Appendix F. □

11 Equivalence principle

Theorem 11.1 (Universal coupling). *The graviton–matter coupling $\kappa h_{\mu\nu} T^{\mu\nu}$ is universal: κ is mode-independent, $\kappa = 1/2$ in natural units.*

Proof. The emergent metric g on $\ker D$ is D_4 -invariant [4], Theorem 7.2. The minimal coupling prescription $\partial_\mu \rightarrow \nabla_\mu$ respects this invariance. By Schur’s lemma, a D_4 -invariant coupling between $B_1 \oplus B_2$ and $V_{14} \otimes V_{14}$ is unique up to overall normalization. Since the graviton is a single field and the coupling is mediated by the same D_4 -invariant tensor, κ is the same for all 14 modes. $\square$

Theorem 11.2 (Equivalence principle). *For each of the 14 modes at $\lambda = 4$:*

$$\frac{Q_{\text{grav}}^{(i)}}{m_{\text{inertial}}^{(i)}} = 1, \quad (30)$$

exactly, with zero D_4 -violating corrections at any order.

Proof. $Q_{\text{grav}}^{(i)} = m_i$ (coefficient of h_{00} in the effective Hamiltonian); $m_{\text{inertial}}^{(i)} = m_i$ (from the dispersion relation $\omega = k^2/(2m_i)$). Both equal m_i , so the ratio is 1. Universality of coupling (Theorem 11.1) guarantees no mode-dependent corrections. $\square$

12 Derivation of λ_L

12.1 The disclosed input

The framework’s mass normalization has been carried as the disclosed input [20, 21]:

$$M_P^2 = N k_L^2 \lambda_L^2, \quad (31)$$

with λ_L described as a residual of the retired SYK/QIN layer.

12.2 The derivation

Theorem 12.1 (λ_L from framework integers). *Using the framework constants $N = L^2$, $kL_{\text{bare}} = (d-1)N/(d+1)$, and the FP measure $8\pi = 2\pi \dim \ker D$, the disclosed input (31) reduces to the consistency relation*

$$\lambda_L = \frac{(d+1) M_P}{(d-1) L^3} = \frac{5 M_P}{1536}. \quad (32)$$

Equivalently, $\lambda_L/M_P = (d+1)/[(d-1)L^3] = 5/1536$.

Proof. From (31): $\lambda_L = M_P/(\sqrt{N} k_L)$. Using $kL_{\text{bare}} = (d-1)N/(d+1)$ and $N = L^2$:

$$\lambda_L = \frac{M_P}{L \cdot (d-1)L^2/(d+1)} = \frac{(d+1)M_P}{(d-1)L^3}. \quad (33)$$

At $d = 4$, $L = 8$: $\lambda_L = 5M_P/1536$. $\square$

Corollary 12.2 (Numerical value). $\lambda_L = 5 \times 1.2209 \times 10^{19}/1536 = 3.974 \times 10^{16}$ GeV.

Corollary 12.3 (Relation to GUT scale). $\lambda_L/M_{\text{GUT}} = (d+1)/(d-1) = 5/3$, where $M_{\text{GUT}} = M_P/L^3 = M_P/512$.

12.3 Interpretation

The disclosed input $M_P^2 = Nk_L^2\lambda_L^2$ is no longer a genuine independent input. It is a consistency relation between M_P , λ_L , and the framework integers N, kL, L . Once the framework integers are fixed and any one mass scale is specified, λ_L is uniquely determined.

The framework now has a single dimensionful anchor: M_P (equivalently v_{EW} , related by $M_P = v_{EW} \exp(kL_{bare})$). All dimensionless ratios in the gravity sector are determined by the substrate geometry.

Remark 12.4 (Philosophical reading). λ_L is the substrate-scale mass — what M_P looks like when viewed through the substrate’s own processing volume $L^3 = 512$, weighted by the ratio $(d+1)/(d-1) = 5/3$ of trivial to non-trivial isotypic content.

13 Predictions and tests

1. $G_N = 25/(8\pi M_P^2)$ exactly (matches CODATA to 0.14%).
2. No deviation from Newton’s law at any observable scale.
3. No dark photon in the Newtonian sector.
4. $O(\epsilon^2)$, $O(\epsilon^4)$, $O(\epsilon^6)$ are the only corrections to Newton’s law.
5. $c_g = c_\gamma$ exactly.
6. No Gribov copies in the HCSM Lorenz gauge.
7. Equivalence principle exact (gravitational charge = inertial mass).
8. Geodesic equation from HCSM (not assumed from GR).
9. $\lambda_L = 5M_P/1536$.

14 Discussion

14.1 What is derived

The full classical Newtonian gravity chain: substrate $\rightarrow$ graviton $\rightarrow$ Einstein–Hilbert $\rightarrow$ linearized $\rightarrow$ Poisson $\rightarrow$ geodesic $\rightarrow \varphi(r) = -G_N M/r$. The geodesic equation and equivalence principle are theorems. λ_L is derived from framework integers.

14.2 Single dimensionful anchor

M_P (equivalently v_{EW}). All dimensionless ratios are framework integers.

14.3 Open items

- First post-Newtonian (1PN) corrections from the full Einstein–Hilbert action — computable, not yet computed.
- Complete non-Abelian gauge-fixing proof — sketch given, full proof open.

14.4 Comparison with standard GR

HCSM reproduces standard Newtonian gravity. Differences appear only at $O(\epsilon^2)$. HCSM has fewer empirical anchors than GR: the functional form of G_N is derived, and λ_L is now derived from M_P .

15 Conclusion

We have derived Newtonian gravity as the static, weak-field, non-relativistic limit of HCSM. The graviton is the self-paired $B_1 \oplus B_2$ curvature mode, exactly massless. The effective action is exactly Einstein–Hilbert. The Newton constant is $G_N = 25/(8\pi M_P^2)$. The Newtonian potential is $\varphi(r) = -G_N M/r$; the Poisson equation is $\nabla^2 \varphi = 4\pi G_N \rho_m$. The geodesic equation and equivalence principle are theorems. The $O(\epsilon^2)$, $O(\epsilon^4)$, $O(\epsilon^6)$ corrections are explicit. Gauge-fixing is complete. $c_g = c_\gamma$ exactly. $\lambda_L = 5M_P/1536$.

The framework has a single dimensionful anchor M_P , with all dimensionless ratios determined by the substrate geometry. Physics is what the machine cannot cancel.

A Linearized HCSM field equations

Expand $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ to quadratic order. Einstein–Hilbert action:

$$S_{\text{EH}}^{(2)} = \frac{1}{16\pi G_N} \int d^4x \left[-\frac{1}{2} \partial_\lambda h_{\mu\nu} \partial^\lambda h^{\mu\nu} + \frac{1}{2} \partial_\lambda h \partial^\lambda h + \partial_\mu h^{\mu\nu} \partial_\nu h - \partial_\mu h^{\mu\nu} \partial_\lambda h_\nu^\lambda \right]. \quad (34)$$

Add harmonic gauge fixing $S_{\text{GF}} = -(1/32\pi G_N) \int d^4x (\partial^\mu \bar{h}_{\mu\nu})^2$. Variation gives $\square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu}$.

B Non-relativistic limit

For static source, $\square \rightarrow \nabla^2$, so $\nabla^2 \bar{h}_{00} = -16\pi G_N \rho_m$. Using $\bar{h}_{00} \approx -2\varphi$, $\nabla^2 \varphi = 4\pi G_N \rho_m$. For point mass, $\varphi(r) = -G_N M/r$.

C $O(\epsilon^2)$ correction

Write $\varphi = \varphi_0 + a^2 \varphi_2$. Then $\nabla^2 \varphi_0 = 4\pi G_N M \delta^{(3)}(x) \Rightarrow \varphi_0 = -G_N M/r$. Using $\sum_\mu \partial_\mu^4 (1/r) = -4/r^5$, $\nabla^2 \varphi_2 = -G_N M/(3r^5)$, giving $\varphi_2 = -G_N M/(18r^3)$. Hence $\varphi = -(G_N M/r)[1 + \epsilon^2/18 + O(\epsilon^4)]$.

D $O(\epsilon^4)$ correction

Using $\sum_\mu \partial_\mu^6 (1/r) = -48/r^7$ and $\sum_\mu \partial_\mu^4 (1/r^3) = 72/r^7$:

$$\nabla^2 \varphi_4 = \frac{G_N M}{3r^7} - \frac{2G_N M}{15r^7} = \frac{G_N M}{5r^7}, \quad (35)$$

giving $\varphi_4 = G_N M/(100r^5)$. Hence $\varphi = -(G_N M/r)[1 + \epsilon^2/18 - \epsilon^4/100 + O(\epsilon^6)]$.

E $O(\epsilon^6)$ correction

Using $\sum_\mu \partial_\mu^8 (1/r) = -576/r^9$, $\sum_\mu \partial_\mu^6 (1/r^3) = 864/r^9$, $\sum_\mu \partial_\mu^4 (1/r^5) = 504/r^9$:

$$\nabla^2 \varphi_6 = -\frac{21}{50} \frac{G_N M}{r^9} + \frac{2}{15} \frac{G_N M}{r^9} - \frac{1}{35} \frac{G_N M}{r^9} = -\frac{331}{1050} \frac{G_N M}{r^9}. \quad (36)$$

Using $\nabla^2(r^{-7}) = 42r^{-9}$, $\varphi_6 = -(331/44100)G_N M r^{-7}$. Hence the full result (21).

F Geodesic equation derivation

From the DNLS action (23), minimal coupling, and eikonal approximation, Hamilton–Jacobi:

$$\frac{dx_0^i}{dt} = \frac{k_i}{m_i}, \quad \frac{dk_i}{dt} = -m_i \partial^i \varphi. \quad (37)$$

Differentiating: $d^2 x_0^i / dt^2 = -\partial^i \varphi$. The relativistic generalization gives the geodesic equation $d^2 x^\mu / d\tau^2 + \Gamma_{\nu\rho}^\mu \dot{x}^\nu \dot{x}^\rho = 0$.

G Reproducible verification

All numerical claims verified at machine precision in IEEE-754 double precision. Script `12-VERIFICATION/HCSM-62_verification.py` verifies: graviton mode count (108); Schur orthogonality; self-coupling counts; bilinear trace 25; 8π identity; exact EH action; linearized equations; Poisson equation; $\varphi(r)$; coefficients $1/18$, $-1/100$, $-331/44100$; explicit $\rho_m(x)$; gauge-fixing completeness; $c_g = c_\gamma$; $\lambda_L = 5M_P/1536$; geodesic equation; equivalence principle.

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